

EXCEL BASICS

A Friendly, Practice-First Study Guide

Section 1 · Understanding the Basics

Who this is for

Absolute beginners, students prepping for a class or interview, and anyone who wants a no-fluff refresher before opening a spreadsheet. Read a section, try the example in real Excel, then attempt the practice task before moving on.

How to use this guide

- Every concept follows the same rhythm: plain-English explanation → example → "why it matters" → quick practice.
- Grey boxes are code / formulas exactly as you'd type them into Excel.
- Coloured callouts are shortcuts, warnings, and memory tricks — the things people actually forget.
- Tables at the end of each part are self-check questions. Try answering from memory first.

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0. The Big Picture — How Excel Is Organised

Before touching a single formula, it helps to picture Excel the way you'd picture a filing cabinet. A cabinet (workbook) holds folders (worksheets), each folder holds pages of ruled paper (rows and columns), and every little box on that paper (a cell) can hold a number, a word, or a formula that calculates something.

Level	Real-world analogy	Excel term
1	The whole filing cabinet	Workbook (the .xlsx file)
2	One folder inside the cabinet	Worksheet (a tab at the bottom)
3	A ruled page inside the folder	Rows + Columns
4	One box on the page	Cell
5	What's written in the box	Data or Formula

Quick example

A workbook called Sales_Report.xlsx might contain three worksheets: Sales, Employees, and Summary. Inside the Sales worksheet, the cell where column B meets row 5 is simply called B5 — that's its address.

Column B + Row 5 = Cell B5

Memory trick

Cell address = Column letter first, then Row number. Always. ("B5", never "5B".)

1. Getting Comfortable With the Excel Screen

Open a blank workbook and look around before you type anything. Every part of the screen has a job, and knowing the names saves a lot of confusion later when a tutorial says "click the ribbon" and you're not sure what that even is.

Area	What it actually does
Title Bar	Shows the file name at the very top
Quick Access Toolbar	Tiny row of your favourite one-click commands
Ribbon	The big toolbar strip — where most commands live
Tabs	Home, Insert, Page Layout, Formulas, Data, Review, View
Name Box	Tells you WHERE you are (e.g. B5)
Formula Bar	Tells you WHAT is in that cell
Column / Row Headers	The A-B-C letters and 1-2-3 numbers
Worksheet Area	The actual grid you work in

Area	What it actually does
Sheet Tabs	Switch between worksheets at the bottom
Status Bar	Quick stats (sum, average, count) for whatever's selected

Remember it this way

Name Box = WHERE you are. Formula Bar = WHAT is there. Click B5 and the Name Box says "B5" while the Formula Bar shows its contents.

2. Workbook vs Worksheet — Don't Mix These Up

This trips up almost every beginner in an interview or a viva, so it's worth nailing down early.

Term	Plain-English meaning
Workbook	The entire file, e.g. Sales_Report.xlsx — think of it as the whole book.
Worksheet	One page/tab inside that file, e.g. "Sales", "Employees", "Summary" — think of it as one page in the book.

One-liner to remember

Workbook = Book. Worksheet = Page. One book can have many pages.

3. Ribbons, Tabs, Groups & Menus

The Ribbon is the wide strip of icons at the top. It's organised in three layers, and once you see the pattern, finding any command becomes second nature.

TAB → GROUP → COMMAND

Example: Home → Font → Bold

So when someone says "go to Home, then look in the Font group, then click Bold" — that's just three clicks down the same hierarchy.

Common tabs you'll use daily

- Home — formatting, fonts, basic editing
- Insert — charts, tables, pictures
- Page Layout — margins, print area
- Formulas — function library, name manager
- Data — sorting, filtering, importing data
- Review — spell check, comments, protection

- View — freeze panes, zoom, gridlines

4. Customising the Quick Access Toolbar (QAT)

The QAT is that tiny strip of icons near the very top-left — usually Save, Undo, and Redo by default. If you find yourself repeating the same command constantly, pin it here so it's always one click away instead of buried in a tab.

Remember it this way

QAT = Quick Access. It exists purely to save you clicks.

5. Cells, Rows & Columns

This is the real foundation of Excel — everything else is built on top of these three ideas.

Concept	Direction / shape	Example
Row	Runs horizontally →	1, 2, 3, 4 ...
Column	Runs vertically ↓	A, B, C, D ...
Cell	Where a row meets a column	C8 = Column C, Row 8

5.1 Naming Cells / Ranges

Typing `=SUM(B2:B20)` works fine, but it's not very readable six months later. Excel lets you give a range a memorable name instead.

Instead of: `=SUM(B2:B20)`
 Use a name: `=SUM(SalesData)`

- Good names: SalesData, EmployeeList, TaxRate
- Avoid vague names like Range1 or Data2 — they don't tell you anything six months from now.

5.2 Selecting Cells

What you want to select	How to do it
A single cell	Click it
Adjacent cells	Click and drag (e.g. A1:A10)
An entire row	Click the row number
An entire column	Click the column letter
Non-adjacent cells	Hold Ctrl while clicking each one

5.3 Formatting Cells

Formatting only changes how a value looks, not the value itself. A cell can still mathematically be 0.25 while displaying as 25%.

Category	Common options
Font	Bold, Italic, Underline, size, typeface
Alignment	Left / Center / Right, Top / Middle / Bottom
Number format	General, Number, Currency, Accounting, Percentage, Date, Time

Raw value: 0.25
With % formatting: 25%

5.4 Auto-Adjusting Rows & Columns

If text is spilling out of a cell or getting cut off, don't manually drag the column wider — double-click the little boundary line between two column headers and Excel resizes it perfectly to fit the content.

Remember it this way

Double-click the boundary = instant AutoFit. Works for row heights too.

5.5 Creating & Deleting

You can insert or delete cells, rows, columns, and entire worksheets. The one thing that catches people out:

Important distinction

Pressing the Delete key only clears a cell's contents — the cell itself stays put. Using "Delete Cells" from the menu actually removes the cell and shifts everything around it.

6. Saving, Undo & Redo

Action	Shortcut	What it does
Save	Ctrl + S	Saves the workbook — use this constantly
Undo	Ctrl + Z	Reverses your last action
Redo	Ctrl + Y	Re-applies an action you just undid

Remember it this way

Ctrl+Z = go back. Ctrl+Y = go forward. Think of it as rewind and fast-forward.

7. Getting Around Fast (Intelligent Navigation)

Manually scrolling through a 1,000-row sheet is painfully slow. These shortcuts are what separate someone comfortable in Excel from someone still learning.

Shortcut	Action
Arrow keys	Move one cell at a time
Enter	Move down one cell
Tab	Move right one cell
Ctrl + Arrow	Jump to the edge of a block of data
Home	Jump to the start of the current row
Ctrl + Home	Jump straight to cell A1
Ctrl + End	Jump to the last used cell in the sheet
Page Up / Page Down	Move one screen up or down
F5 or Ctrl + G	Open "Go To" and jump to any cell by typing its address

Example: if data fills A1:A1000, click A1 and press Ctrl + ↓ — Excel jumps you straight to the bottom of that block instead of making you scroll for ten seconds.

8. Selecting Ranges

A range is simply a group of cells treated as one block, written as start:end.

Notation	Meaning
A1	A single cell
A1:A10	One column, 10 rows
A1:C1	One row, 3 columns
A1:C10	A rectangle — 3 columns × 10 rows

Shortcut	Selects
Ctrl + Space	The entire column
Shift + Space	The entire row
Ctrl + A	Everything in the sheet

9. Copying & Pasting Mastery

Action	Shortcut	What happens to the original
Copy	Ctrl + C	Stays exactly where it was
Cut	Ctrl + X	Removed once pasted elsewhere
Paste	Ctrl + V	Places the copied/cut content
Undo	Ctrl + Z	Reverses the last paste/cut

Copy vs Cut, visually

Copy: A1 → Ctrl+C → paste in B1 → A1 stays, B1 is a duplicate
 Cut: A1 → Ctrl+X → paste in B1 → A1 is empty, B1 has the value

Paste Special

Sometimes you only want part of what you copied — say, the calculated number from a formula, not the formula itself. Paste Special lets you choose exactly what sticks: Values only, Formulas only, Formatting only, or just the column widths.

Common use case

A1 contains a formula. You want just the result in another cell without the formula tagging along — use Paste Special → Values.

10. Basic Arithmetic Operations

Every Excel formula begins with an equals sign. Without it, Excel just treats your entry as plain text.

Operator	Meaning	Example	Result
+	Addition	=10+5	15
-	Subtraction	=10-5	5
*	Multiplication	=10*5	50
/	Division	=10/5	2
^	Power	=10^2	100
%	Percentage	=10%	0.1

10.1 Using Cell References Instead of Hard-Coded Numbers

Typing =100+50 gives you an answer, but it's frozen in time. Reference the cells instead, and the formula updates automatically the moment the underlying numbers change.

A	B	C
100	50	=A1+B1 → 150

Why it matters

This is the entire point of a spreadsheet: change one input, and every formula depending on it recalculates instantly.

10.2 Built-in Functions You'll Use Constantly

Function	What it returns	Example
SUM	Total of a range	=SUM(A1:A10)
AVERAGE	Mean of a range	=AVERAGE(A1:A10)
MIN	Smallest value	=MIN(A1:A10)
MAX	Largest value	=MAX(A1:A10)
COUNT	How many cells contain numbers	=COUNT(A1:A10)
COUNTA	How many cells are non-empty	=COUNTA(A1:A10)

11. Cell Referencing — Relative, Absolute & Mixed

This is the single most important concept in the whole beginner syllabus, and the one most people never fully lock in. Understand this properly and copying formulas stops feeling unpredictable.

11.1 Relative Reference — "Moves With You"

A plain reference like A1 shifts automatically when you copy the formula elsewhere.

=A1+B1 copied one row down becomes =A2+B2

11.2 Absolute Reference — "Locked in Place"

Add dollar signs and the reference never changes, no matter where you copy the formula.

=\$A\$1 copied anywhere still reads =\$A\$1

11.3 Mixed Reference — "Half-Locked"

Style	What's locked	What can move
\$A1	Column (A)	Row can change
A\$1	Row (1)	Column can change

11.4 The Dollar Sign as a "Lock"

Notation	Column locked?	Row locked?
A1	No	No
\$A1	Yes	No
A\$1	No	Yes
\$A\$1	Yes	Yes

Memory trick

\$ means LOCKED. Whatever comes right after the \$ can't move when you copy the formula.

11.5 Worked Example — Tax Calculation

Say A2 is a product's price, B2 is the quantity sold, and the tax rate lives in a single fixed cell, C1 (18%). You want to copy this formula down for every product without the tax rate reference drifting.

```
=A2*B2*$C$1
```

The price and quantity references (A2, B2) are left relative because they should change for every row. The tax rate (\$C\$1) is locked because every row needs to point at that exact same cell.

Visual: how far each reference type "travels" when copied

Reference type	Relative freedom to move	Locked
A1		0%
\$A1		col
A\$1		row
\$A\$1		100%

12. Techniques for Filling Data

12.1 The Fill Handle

That tiny square sitting at the bottom-right corner of a selected cell is one of the most time-saving tools in Excel. Drag it and Excel extends a pattern for you.

Type: 1, 2, 3
 Drag the fill handle down
 Result: 4, 5, 6 ...

12.2 Copying a Formula Down

Because references are relative by default, dragging a formula down automatically updates row numbers for you.

A	B	C
100	20	=A2+B2
...	...	=A3+B3 (auto-adjusted)
...	...	=A4+B4 (auto-adjusted)

12.3 Filling a Series

The Fill Handle isn't limited to plain numbers — it recognises common patterns automatically.

Start typing	Excel auto-continues with
1, 2	3, 4, 5 ...
10, 20	30, 40, 50 ...
Monday	Tuesday, Wednesday, Thursday ...
01-Jan	02-Jan, 03-Jan, 04-Jan ...

12.4 Double-Click the Fill Handle

If there's an adjacent column already full of data, double-clicking the fill handle (instead of dragging) fills your formula all the way down to match — perfect for large datasets where dragging manually would take forever.

12.5 Keyboard Fill Shortcuts

Shortcut	Action
Ctrl + D	Fill Down — copies the top cell downward
Ctrl + R	Fill Right — copies the left cell rightward

13. The Master Shortcut Table

Bookmark this page. These cover roughly 90% of everyday Excel work.

Shortcut	Purpose
Ctrl + S	Save
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + C	Copy
Ctrl + X	Cut

Shortcut	Purpose
Ctrl + V	Paste
Ctrl + A	Select all
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + G / F5	Go To
Ctrl + Home	Go to A1
Ctrl + End	Go to last used cell
Ctrl + Arrow	Jump through a block of data
Ctrl + D	Fill Down
Ctrl + R	Fill Right
Ctrl + Space	Select entire column
Shift + Space	Select entire row
F2	Edit the active cell
Alt + Enter	Start a new line inside a cell
Ctrl + ;	Insert today's date
Ctrl + Shift + :	Insert current time

14. Five Formula Rules to Never Break

#	Rule	Example
1	Every formula starts with =	=A1+B1
2	Prefer cell references over hard-coded numbers	=A1+B1, not =100+50
3	Use \$ to lock a reference that must stay fixed	=\$A\$1
4	Functions always use parentheses	=SUM(A1:A10)
5	A range uses a colon	A1:A10 means A1 through A10

15. Common Beginner Mistakes (and the Fix)

Mistake	Wrong	Correct
Forgetting the equals sign	A1+B1	=A1+B1
Using x instead of *	A1 x B1	=A1*B1

Mistake	Wrong	Correct
Wrong reference type when copying	=A2*B1	=A2*\$B\$1 (if B1 must stay fixed)
Confusing Delete key with deleting the cell	—	Delete key clears contents only; "Delete Cells" removes structure
Typing repetitive data by hand	1,2,3,4,5 typed one by one	Use the Fill Handle / Fill Series

16. Quick Memory Map

If you can walk through this chain from memory, you understand the whole first section.

LAYOUT → WORKBOOK/WORKSHEET → CELLS/ROWS/COLUMNS → SELECT → FORMAT
→ COPY/PASTE → FORMULAS → CELL REFERENCES → FILL

Concept	Symbol / Shortcut
Cell	A1
Range	A1:C10
Formula	=A1+B1
Function	=SUM(A1:A10)
Relative	A1
Absolute	\$A\$1
Mixed	\$A1 or A\$1
Fill Down	Ctrl + D
Fill Right	Ctrl + R

17. Hands-On Practice — Retail Order Sheet

Open a blank workbook and build this exact table. Put the tax rate in its own cell so you can practice locking a reference.

Product	Price	Quantity	Tax Rate	Total
Laptop	50000	2	18%	
Mouse	800	5	18%	
Keyboard	1500	3	18%	
Monitor	12000	2	18%	
Headphones	2500	4	18%	

Put the master tax rate in cell G1 (18%), then build the Total column in two stages:

Stage 1: `=B2*C2`
 Stage 2: `=B2*C2*$G$1`

Copy the formula down for every product and watch what happens to each part of the reference.

What to notice

B2 and C2 shift automatically as you copy down (relative), while \$G\$1 stays glued to the same cell (absolute) — exactly the tax scenario from Section 11.5.

18. Practice Challenge — Level 1 (Foundations)

Task	What to practice
1	Create a list from 1–20 using the Fill Handle
2	AutoFill the days Monday → Sunday
3	Build Price × Quantity = Total using cell references
4	Calculate Total, Average, Minimum, and Maximum with functions
5	Copy your formulas down using the Fill Handle
6	Use Ctrl + D to fill formulas instead of dragging
7	Set a fixed tax rate and reference it with \$G\$1 style locking

19. Practice Challenge — Level 2 (Salary Sheet)

Employee	Basic Salary	Bonus %	Bonus	Final Salary
Rahul	25000	10%		
Priya	30000	12%		
Amit	28000	8%		
Sneha	35000	15%		
Rohan	40000	10%		

Your checklist:

#	Task
1	Calculate Bonus = Basic Salary × Bonus %
2	Calculate Final Salary = Basic Salary + Bonus
3	Use formulas, never manual arithmetic
4	Copy formulas down for every employee
5	Calculate the average salary
6	Find the minimum salary
7	Find the maximum salary
8	Format the salary columns as Currency
9	Format the Bonus % column as Percentage
10	AutoFit every column

20. Practice Challenge — Level 3 (Mini Sales Report)

Build a report with these columns: Date, Product, Salesperson, Quantity, Price, Revenue, Tax, Final Revenue.

Formula	Logic
Revenue	Quantity × Price
Tax	Revenue × a fixed tax-rate cell (absolute reference)
Final Revenue	Revenue + Tax

Then calculate Total Revenue, Average Revenue, Highest Revenue, and Lowest Revenue using SUM, AVERAGE, MAX, and MIN.

Final polish checklist:

- Format the table (currency, percentages, borders)

- AutoFit all columns
- Use the Fill Handle for repetitive formulas
- Use absolute references for the tax rate
- Save the workbook (Ctrl + S)
- Practice Undo/Redo on purpose, just to see it work
- Navigate the sheet using only keyboard shortcuts — no mouse

21. Self-Check — Can You Explain These Without Looking Back?

Cover the earlier sections and try answering each of these out loud or on paper. If you get stuck, that's exactly the spot to re-read.

Basics

#	Question
1	What is a workbook, and how is it different from a worksheet?
2	What is a cell? What is a range?
3	Which direction do rows run? Which direction do columns run?

Navigation

#	Question
1	How do you jump straight to cell A1?
2	How do you jump to the end of a dataset?
3	How do you select an entire row? An entire column?

Formulas & Functions

#	Question
1	What symbol must every formula start with?
2	Which operator multiplies two cells?
3	What do SUM, AVERAGE, and COUNT each return?

References

#	Question
1	What's the difference between a relative and an absolute reference?
2	What does \$A\$1 mean? What about \$A1 and A\$1?

Fill

#	Question
1	What is the Fill Handle, and where do you find it?
2	What do Ctrl + D and Ctrl + R each do?

22. One-Minute Revision (Read This Right Before a Test)

Workbook = the Excel file Worksheet = a page inside it
 Cell = Column + Row (e.g. B5) Range = a group of cells (e.g. A1:C10)

Formula always starts with =
 SUM = total AVERAGE = mean MIN = smallest MAX = largest COUNT = numbers

Relative (A1) → moves when copied
 Absolute (\$A\$1) → locked, never moves
 Mixed (\$A1 / A\$1) → column or row locked, not both

Copy = Ctrl+C Paste = Ctrl+V Cut = Ctrl+X
 Undo = Ctrl+Z Redo = Ctrl+Y
 Fill Down = Ctrl+D Fill Right = Ctrl+R
 Select Column = Ctrl+Space Select Row = Shift+Space
 Go to A1 = Ctrl+Home Last used cell = Ctrl+End

23. The Golden Rule

“Don't just read Excel. Build something with it.”

LEARN → SEE AN EXAMPLE → DO IT YOURSELF → MAKE A MISTAKE → FIX IT → REPEAT

If you can explain a concept, perform it live in a spreadsheet, and apply it to a small dataset of your own, you actually know it — not just recognise it.

Section 1 Completion Checklist

Done?	Skill
☐	Understand the Excel layout
☐	Understand workbook vs worksheet
☐	Understand ribbons, tabs, and groups
☐	Customise the Quick Access Toolbar
☐	Understand cells, rows, and columns
☐	Select cells and ranges confidently
☐	Format cells (font, alignment, number formats)
☐	AutoFit rows and columns
☐	Create and delete cells/rows/columns
☐	Save, Undo, and Redo
☐	Navigate Excel using keyboard shortcuts
☐	Copy, cut, and paste — including Paste Special
☐	Perform basic arithmetic with formulas
☐	Use SUM, AVERAGE, MIN, MAX, COUNT, COUNTA
☐	Understand relative, absolute, and mixed references
☐	Use the Fill Handle, Fill Down, and Fill Right
☐	Complete Practice Levels 1, 2, and 3

You're done with Section 1 when...

You can open a completely blank workbook and independently build a small, formatted, formula-driven dataset — without following a tutorial step by step.